

Vectors: Lists of Numbers as Arrows

The basic building block of every LLM · video 1:56

This lesson has one big idea: a **vector** is a **list of numbers** that you can also picture as an **arrow**. You can add vectors, stretch them and measure their length, and the rules are the same whether a vector has 2 numbers or 768. Every token inside an LLM is represented by one.

How to use this guide: master one concept before moving to the next.

1. Watch the whole video once, without stopping.
2. Then take the concepts **in order**. For each one: read the short explanation, look at the picture from the video (▶ is the time in the video), and answer every question in its *Check* without looking anything up.
3. Check your answers in the **Answers** section at the end (page 10).
4. **Pass mark: every question in the check correct.** If you missed one, rewatch that concept's video segment, reread the explanation, and redo only the questions you missed. Tick the concept off below, then move on.

It is strongly suggested that you **master each concept before starting the next one**. Each concept builds on the previous ones.

Concepts in this lesson

CONCEPT	VIDEO SEGMENT	QUESTIONS	MASTERED
1. A vector is a list of numbers	▶ 0:08-0:27	3	☐
2. Two numbers, one arrow	▶ 0:27-0:51	4	☐
3. Adding vectors	▶ 0:51-1:02	4	☐
4. Scaling: stretch, shrink, flip	▶ 1:02-1:13	4	☐
5. Length, by Pythagoras	▶ 1:13-1:21	4	☐
6. Directions that mean something	▶ 1:21-1:32	3	☐
7. Vectors in NumPy	▶ 1:32-1:54	3	☐

Before you start: this is the first Foundations video, so nothing earlier is needed: adding, multiplying and a square root are enough. Vectors are used throughout the main series, especially in episodes 1 (What is an LLM?), 3 (Embeddings), 4 (Position) and 9 (The Transformer Block). Concept 7 reads a few lines of NumPy; F11 (NumPy in Three Minutes) covers more.

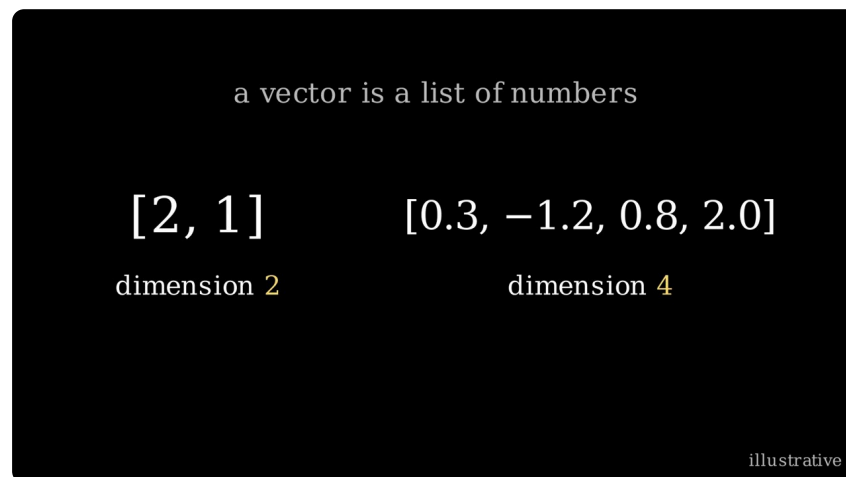
CONCEPT 1 OF 7

A vector is a list of numbers

▶ Video 0:08–0:27

Everything an LLM does, it does with vectors: every token, every hidden state and every prediction starts as one. At its simplest, a **vector** is just a **list of numbers**, such as $[2, 1]$. The **order** matters: each position has its own meaning, so $[2, 1]$ and $[1, 2]$ are different vectors.

How many numbers a vector has is called its **dimension**. $[2, 1]$ has dimension 2; $[0.3, -1.2, 0.8, 2.0]$ has dimension 4.



▶ 0:27 Two vectors: $[2, 1]$ has dimension 2, and $[0.3, -1.2, 0.8, 2.0]$ (illustrative values) has dimension 4.

KEY IDEA

A vector is an **ordered list of numbers**. Its **dimension** is how many numbers it has.

Check 1: A vector is a list of numbers

Answer all 3 questions before looking at the answers. Pass mark: all correct.

1.1 CALCULATION

What is the dimension of each vector? (a) $[5, 0, -2]$ (b) $[7]$ (c) $[0, 0, 0, 0, 0, 0]$

.....

1.2 TRUE OR FALSE

" $[2, 1]$ and $[1, 2]$ are the same vector, because they contain the same numbers."

True ☐ False ☐

1.3 SHORT ANSWER

You describe a flat as $[\text{bedrooms}, \text{bathrooms}, \text{floor area in m}^2, \text{year built}] = [3, 2, 95, 1987]$.

What is its dimension, and what does the second number mean?

.....

.....

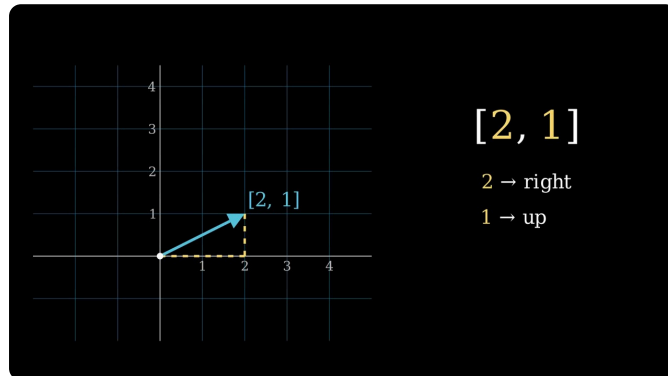
CONCEPT 2 OF 7

Two numbers, one arrow

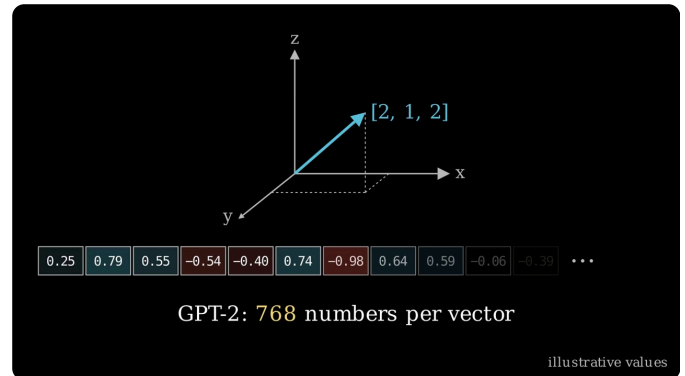
▶ Video 0:27-0:51

With two numbers we can draw a vector. The first number says how far to go **right**, the second how far to go **up**. So $[2, 1]$ is the **arrow from the origin** (the point $[0, 0]$) to the point 2 right, 1 up. Negative numbers go left or down.

Three numbers give an arrow in three-dimensional space. Beyond that we can't draw it, but the calculations work **exactly the same**: adding, scaling and length (the next concepts) don't care how many numbers there are. GPT-2 uses vectors with **768** numbers.



▶ 0:37 $[2, 1]$ as an arrow: 2 to the right, then 1 up, starting at the origin.



▶ 0:50 Three numbers give an arrow in 3-D. GPT-2 uses 768 numbers per vector (values illustrative).

KEY IDEA

A vector is both a **list of numbers** and an **arrow from the origin**. The picture stops at 3 dimensions; the arithmetic doesn't.

Check 2: Two numbers, one arrow

Answer all 4 questions before looking at the answers. Pass mark: all correct.

2.1 SHORT ANSWER

Describe the arrow for $[-3, 2]$: starting at the origin, which way and how far do you go?

2.2 MULTIPLE CHOICE

An arrow starts at the origin and ends 1 to the right and 4 down. Which vector is it?

- A. $[4, 1]$
- B. $[1, 4]$
- C. $[1, -4]$
- D. $[-1, 4]$

2.3 TRUE OR FALSE

"Vectors with 768 numbers can't be drawn, so the rules for adding them are different."

True ☐ False ☐

2.4 CALCULATION

A model uses one vector of dimension 768 per token. How many numbers does it need for the 5 tokens of "The cat sat on the"?

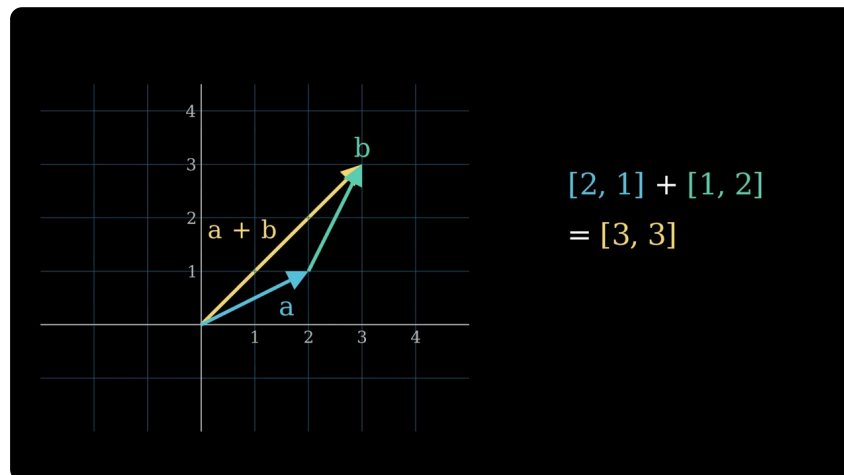
CONCEPT 3 OF 7

Adding vectors

▶ Video 0:51-1:02

To add two vectors, add their numbers **position by position**: $[2, 1] + [1, 2] = [2 + 1, 1 + 2] = [3, 3]$. Both vectors need the same dimension, so that every number has a partner.

As arrows, adding means **placing one arrow at the tip of the other**. The sum is the arrow from the origin to the tip of the second one: walk along a , then along b , and $a + b$ is the shortcut from start to finish.



▶ 1:01 $[2, 1] + [1, 2] = [3, 3]$. Put b at the tip of a ; the sum $a + b$ goes from the origin to where b now ends.

KEY IDEA

Add vectors **position by position**. As arrows: **tip to tail**, and the sum goes from the start of the first to the end of the second.

Check 3: Adding vectors

Answer all 4 questions before looking at the answers. Pass mark: all correct.

3.1 CALCULATION

Compute $[4, -1, 2] + [1, 3, -2]$.

.....

3.2 CALCULATION

Start at the origin, walk along $[2, 1]$, then along $[-3, 2]$. Where do you end up?

.....

3.3 TRUE OR FALSE

" $[1, 2] + [2, 1]$ gives a different result from $[2, 1] + [1, 2]$."

True ☐ False ☐

3.4 MULTIPLE CHOICE

Why can't you add $[1, 2]$ and $[1, 2, 3]$?

- A. Because the result would be negative
- B. Because the third number has no partner: both vectors must have the same dimension
- C. Because only vectors of dimension 2 can be added
- D. You can: the answer is $[2, 4, 3]$

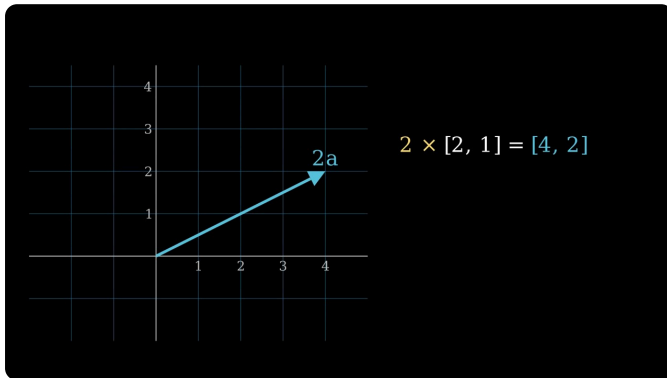
CONCEPT 4 OF 7

Scaling: stretch, shrink, flip

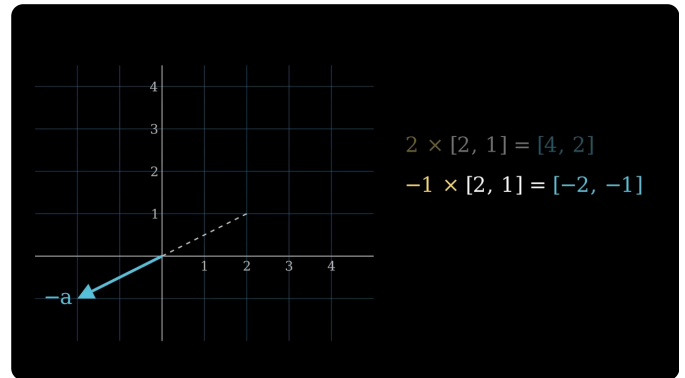
▶ Video 1:02-1:13

Multiplying a vector by a single number, called a **scalar**, multiplies **every** number in it: $2 \times [2, 1] = [4, 2]$. The arrow keeps its direction and becomes twice as long.

A scalar between 0 and 1 shrinks the arrow: $0.5 \times [4, 2] = [2, 1]$. A **negative** scalar flips it around, so it points the opposite way: $-1 \times [2, 1] = [-2, -1]$.



▶ 1:10 $2 \times [2, 1] = [4, 2]$: the same direction, twice as long.



▶ 1:13 $-1 \times [2, 1] = [-2, -1]$: a negative number flips the arrow around.

KEY IDEA

A scalar multiplies **every** number of the vector. The arrow stretches or shrinks along the same line; a **negative** scalar also reverses it.

Check 4: Scaling: stretch, shrink, flip

Answer all 4 questions before looking at the answers. Pass mark: all correct.

4.1 CALCULATION

Compute $3 \times [1, -2, 0]$.

.....

4.2 MULTIPLE CHOICE

Which vector points in exactly the opposite direction to $[2, 1]$?

- A. $[1, 2]$
- B. $[-2, 1]$
- C. $[-4, -2]$
- D. $[2, -1]$

4.3 TRUE OR FALSE

" $0.5 \times [4, 2]$ points the same way as $[4, 2]$, but is half as long."

True ☐ False ☐

4.4 CALCULATION

Compute $2 \times [1, 3] + [-1, 0]$.

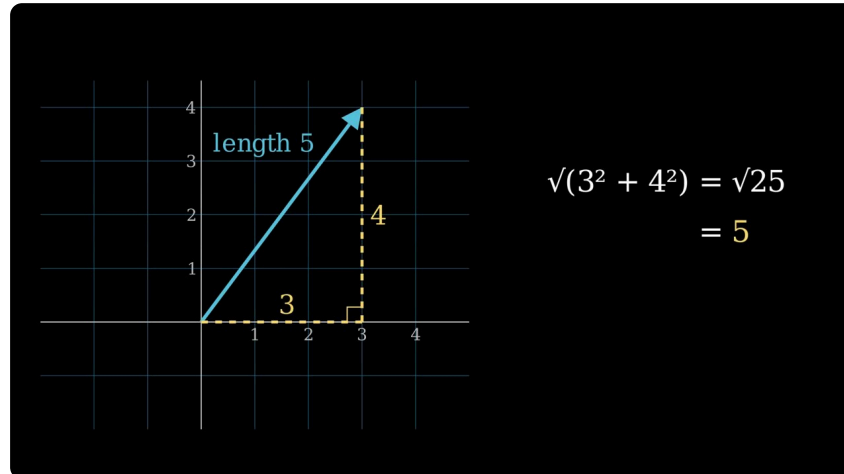
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CONCEPT 5 OF 7

Length, by Pythagoras

▶ Video 1:13–1:21

The **length** of a vector comes from Pythagoras: **square each number, add them up, and take the square root**. For $[3, 4]$: $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. The length of a vector a is often written $\|a\|$; you'll see this in F02.



▶ 1:21 The legs 3 and 4 form a right triangle, so the arrow $[3, 4]$ has length $\sqrt{3^2 + 4^2} = \sqrt{25} = 5$.

The same recipe works in any dimension, with more numbers to square. For $[2, 3, 6]$: $\sqrt{4 + 9 + 36} = \sqrt{49} = 7$. Squares are never negative, so neither is a length.

KEY IDEA

Length = the **square root of the sum of the squares** of the numbers. It works in any number of dimensions.

Check 5: Length, by Pythagoras

Answer all 4 questions before looking at the answers. Pass mark: all correct.

5.1 CALCULATION

What is the length of $[6, 8]$?

.....

5.2 CALCULATION

What is the length of $[1, 1]$, to 3 decimal places?

.....

5.3 CALCULATION

The length of $[3, 4]$ is 5. Without squaring anything, what is the length of $-2 \times [3, 4]$?

.....

5.4 MULTIPLE CHOICE

Which vector has length exactly 3?

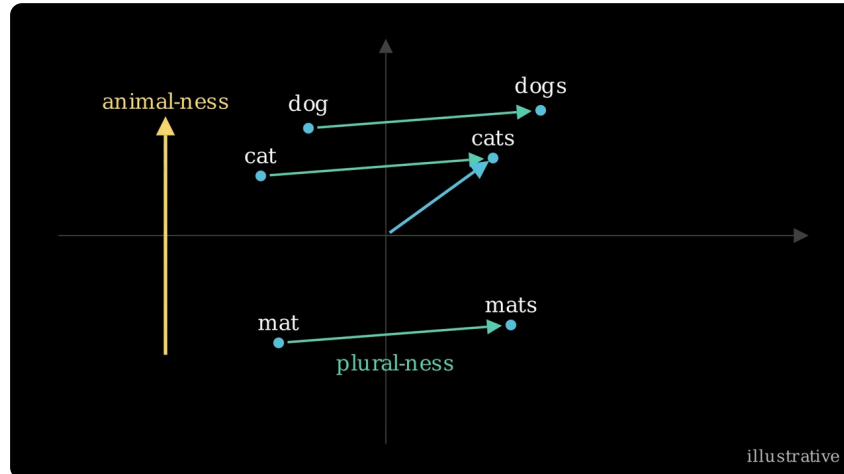
- A. $[1, 2]$
- B. $[3, 3]$
- C. $[2, 1, 2]$
- D. $[1, 1, 1]$

CONCEPT 6 OF 7

Directions that mean something

▶ Video 1:21-1:32

In an LLM, **directions** can come to mean something. One direction might lean toward animals, another toward plurals. A token's vector is a **point in this space of meaning**, and moving along the “plural” direction takes *cat* toward *cats*.



▶ 1:32 An illustrative space of meaning: one direction leans toward animals, another toward plurals. The same “plural” arrow links *cat* → *cats*, *dog* → *dogs* and *mat* → *mats*.

The picture is illustrative. A real model learns its directions from data, in hundreds of dimensions, and they are usually not as neat as one arrow per idea. Episode 3 (Embeddings) looks at this properly.

KEY IDEA

A token's vector is a **point in a space of meaning**. Directions in that space can stand for ideas such as “animal” or “plural”.

Check 6: Directions that mean something

Answer all 3 questions before looking at the answers. Pass mark: all correct.

6.1 CALCULATION

In a toy 2-D space, *cat* = [1, 3] and *cats* = [4, 3]. (a) Which vector must you add to *cat* to get *cats*? (b) *dog* = [2, 5]. If the same “plural” arrow applies, where do you expect *dogs*?

6.2 TRUE OR FALSE

“Each direction in an LLM's vectors was given its meaning by hand, e.g. an engineer decided which direction means ‘animal’.”

True ☐ False ☐

6.3 MULTIPLE CHOICE

A token's vector is best thought of as...

- A. a point (or arrow) in a space where directions can carry meaning
- B. the dictionary definition of the word, stored as text
- C. the token's position in the sentence
- D. a probability for every word in the vocabulary

CONCEPT 7 OF 7

Vectors in NumPy

▶ Video 1:32-1:54

In NumPy, a vector is an **array**: `np.array([2, 1])`. The operations of this lesson take one line each: `a + b` adds position by position, `2 * a` scales, and `np.linalg.norm(a)` gives the length.

```
1 import numpy as np
2 a = np.array([2, 1])
3 b = np.array([1, 2])
4 a + b                                # array([3, 3])
5 2 * a                                # array([4, 2])
6 np.linalg.norm(np.array([3, 4]))    # 5.0
```

▶ 1:37 The NumPy code from the video: adding, scaling and measuring length take one line each.

```
import numpy as np
a = np.array([2, 1])
b = np.array([1, 2])
a + b                                # array([3, 3])
2 * a                                # array([4, 2])
np.linalg.norm(np.array([3, 4]))    # 5.0
```

You'll meet vectors in almost every episode: as **embeddings** in episode 3, **added to positions** in episode 4, and flowing along the **residual stream** in episode 9.

KEY IDEA

A vector is a NumPy array. `+`, `*` and `np.linalg.norm` do the work, for 2 numbers or 768.

Check 7: Vectors in NumPy

Answer all 3 questions before looking at the answers. Pass mark: all correct.

7.1 SHORT ANSWER

With `a` and `b` as above, what does `a + 3 * b` give?

7.2 MULTIPLE CHOICE

What does `np.linalg.norm(np.array([2, 1, 2]))` return?

- A. 5.0
- B. 3.0
- C. 9.0
- D. `array([4, 1, 4])`

7.3 HANDS-ON CODE

Try it yourself. (a) Complete `length(v)` without using `np.linalg.norm`: use `v ** 2` (square every entry), `np.sum` and `np.sqrt`. (b) Predict what the three `print` lines show, then run it. (c) Now try `a + np.array([1, 2, 3])`. What happens, and why?

```
import numpy as np

a = np.array([2, 1])
b = np.array([1, 2])

def length(v):
    ...                # (a) your code here

print(a + b, 2 * a, -1 * a)
print(length(np.array([3, 4])), np.linalg.norm(np.array([3, 4])))
print(length(2 * a) / length(a))
```

ANSWERS

Answers and explanations

Mark each answer right or wrong. If a concept has any wrong answer, rewatch its segment (▶) and redo the questions you missed before moving on.

1. A vector is a list of numbers · rewatch ▶ 0:08-0:27

1.1 (a) 3 (b) 1 (c) 6. Count the numbers. Zeros and negative numbers count like any other entry.

1.2 False. Order matters: the first position and the second position mean different things. As arrows, one goes 2 right and 1 up, the other 1 right and 2 up.

1.3 Dimension 4; the second number is the number of bathrooms. Four numbers, and each position has a fixed meaning. An LLM's vectors work the same way, except that the meanings are learned, not chosen by hand.

2. Two numbers, one arrow · rewatch ▶ 0:27-0:51

2.1 3 to the left, then 2 up. The first number is horizontal (negative means left), the second is vertical (positive means up).

2.2 C. Right comes first (1), then up/down (4 down = -4). A swaps the order; B and D get the signs wrong.

2.3 False. The picture stops at 3 dimensions, but the rules are exactly the same for any number of entries.

2.4 3,840. 5 vectors $\times$ 768 numbers each = 3,840.

3. Adding vectors · rewatch ▶ 0:51-1:02

3.1 [5, 2, 0]. Position by position: $4 + 1$, $-1 + 3$, $2 + (-2)$.

3.2 [-1, 3]. Walking tip to tail is vector addition: $[2 + (-3), 1 + 2] = [-1, 3]$.

3.3 False. Both are $[3, 3]$. Adding numbers doesn't depend on order, so neither does adding vectors.

3.4 B. Addition pairs up positions. With 2 and 3 numbers there is nothing to pair with the 3 (D just invents a 0).

4. Scaling: stretch, shrink, flip · rewatch ▶ 1:02-1:13

4.1 [3, -6, 0]. Every entry is multiplied by 3, including the 0.

4.2 C. $[-4, -2] = -2 \times [2, 1]$: flipped and twice as long. B and D flip only one number, which gives a different direction, not the opposite one.

4.3 True. It is $[2, 1]$: a positive scalar keeps the direction, and 0.5 halves the length.

4.4 [1, 6]. Scale first: $2 \times [1, 3] = [2, 6]$. Then add: $[2 - 1, 6 + 0] = [1, 6]$.

5. Length, by Pythagoras · rewatch ▶ 1:13-1:21

5.1 10. $\sqrt{36 + 64} = \sqrt{100} = 10$. It is $2 \times [3, 4]$, so it is twice as long as $[3, 4]$.

5.2 1.414. $\sqrt{1 + 1} = \sqrt{2} \approx 1.414$. A length doesn't have to be a whole number.

5.3 10. The -2 flips the arrow and doubles it. Flipping doesn't change length, so it is $2 \times 5 = 10$ (check: $[-6, -8]$ has length $\sqrt{100} = 10$).

5.4 C. $\sqrt{4 + 1 + 4} = \sqrt{9} = 3$. The others: $\sqrt{5} \approx 2.24$, $\sqrt{18} \approx 4.24$, $\sqrt{3} \approx 1.73$.

6. Directions that mean something · rewatch ► 1:21-1:32

6.1 (a) [3, 0] (b) [5, 5]. $[1, 3] + [3, 0] = [4, 3]$. Adding the same plural arrow to dog: $[2 + 3, 5 + 0] = [5, 5]$.

6.2 False. The directions come to mean something through training on data; nobody assigns them by hand.

6.3 A. It is a point in a space of meaning. Position (C) is added separately in episode 4; probabilities (D) are the model's output, not a token's vector.

7. Vectors in NumPy · rewatch ► 1:32-1:54

7.1 array([5, 7]). $3 * b = [3, 6]$, then $[2 + 3, 1 + 6] = [5, 7]$.

7.2 B. $\sqrt{4 + 1 + 4} = \sqrt{9} = 3$. A just adds the numbers; C forgets the square root; D only squares them.

7.3 (b) [3 3] [4 2] [-2 -1], then 5.0 5.0, then 2.0 · (c) a ValueError (a) `def length(v):
return np.sqrt(np.sum(v ** 2))`. (b) Line 1: add, scale, flip. Line 2: your function and NumPy agree on 5.0. Line 3: doubling a vector doubles its length, so the ratio is 2.0. (c) NumPy raises `ValueError: operands could not be broadcast together with shapes (2,) (3,)`: a has 2 numbers and the other array has 3, so the third position has no partner.